RESOURCE ~~WASTE~~ MAP

WORLD

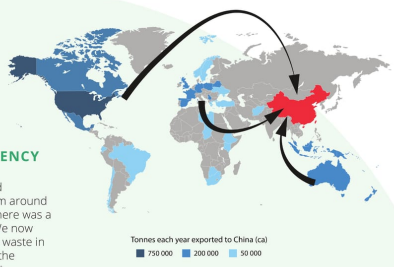


EARTH **OVERSHOOT** DAY

JULY 29th!

Overpopulation and over-use of the earth's resources

The world is not only overpopulated but the world's population is also extracting resources from the earth at a rate faster than it can re-generate. This can easily be illustrated by Earth Overshoot Day - the day in the calendar year when the world has used up all the regenerative resources of that year. In 2019, this was on July 29th! (1)



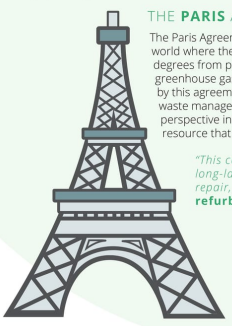
WASTE **EMERGENCY**

When china stopped importing waste from around the world in 2018, there was a waste emergency. We now had to deal with our waste in a new way, and see the consequences of not recycling. (2)

SUSTAINABLE **DEVELOPMENT** GOALS

Protect the Planet

The United Nations members have developed seventeen Sustainable Development Goals (SDGs) for 2030 that focuses on protecting the planet, ending poverty, and improving people lives around the world. (3)



THE **PARIS** AGREEMENT

The Paris Agreement is a contract signed by 186 countries around the world where the goal is to keep the temperature level under 2 degrees from pre-industrial times. The main focus is to reduce greenhouse gases and emotion. If we set out to reach the goals set by this agreement, the world needs to move to a more sustainable waste management system. A key element in this change is the perspective in which waste is viewed. Waste is not waste, but a used resource that has a capacity of re-use and renewal. (4)

"This can be achieved through long-lasting design, maintenance, repair, reuse, re-manufacturing, ***refurbishing and recycling"*** (25)

0% **EMISSION** GOAL

EUROPE



20 European partners, led by research institute ITENE in Valencia, are working on improving the waste treatment for plastic within Europe.

"Recycling of plastic can increase if the rate of ***collection improves*** *and we implement new sorting technology"*

- César Aliaga
Project Coordinator (5)

"Reducing loss of resources from waste management is key to strengthening the circular economy in Europe" (24)

30%
Percentage of plastic waste recycled in Europe

... The number is even lower in Southern- and Eastern Europe

Recycling by the Numbers

EUROPE BY 2035 **65%**

NORWAY TODAY **37%**

475 Kg
Total waste per capita in Europe (6)

#2
Norway has the second most waste per capita in Europe

739 Kg
Total waste per capita in Norway (6)

NORWAY

NORWAY **OVERSHOOT** DAY

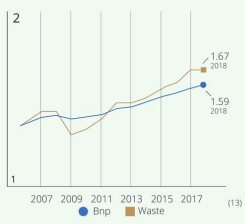
APRIL 18th! (1)



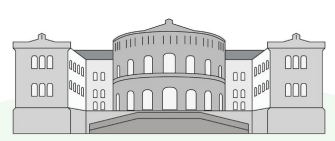
LAGGING **BEHIND**

According to the goal ***Norges miljømål 4.3***, the growth of amount of waste should be a lot lower than the economic growth in the country, however, the Norway's gross national product increased with 2% from 2016 to 2017 while the growth of the amount of waste increased with 3%. These numbers shows that ***Norges miljømål 4.3 goal*** has not been reached. (s40) (7)

Development in **waste compared to GDP.**



(13)



GOVERNMENT REGULATIONS

Business responsibility

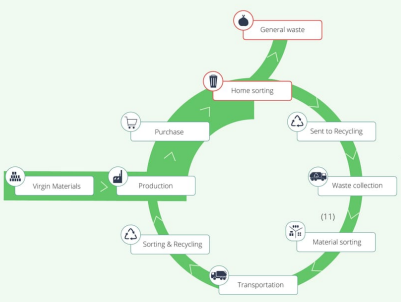
From January 2018, Businesses that produces more than 1000 kg of a certain packaging has to be member of a valid recycling company that is verified by Miljødirektoratet. (8)

Grønt punkt and Norsirk

The packaging producers have to pay a fee and be members of Norsirk or Grønt Punkt. They take care for that the packaging the businesses add to the market is being delivered to recycling and is recycled. For example, Grønt Punkt reports how much waste is delivered to Germany and Germany report back how much is actually recycled. (9)

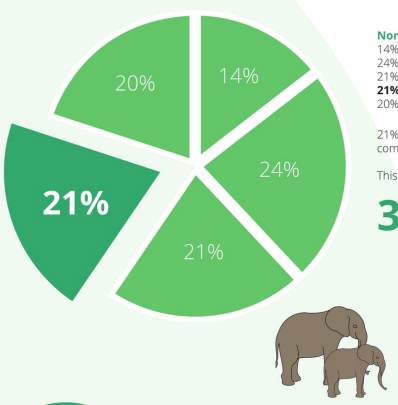
Municipality responsibility

Forurensningsloven states that all the municipalities in Norway have the responsibility for collecting waste generated by the household. In 2017, 39% of all household was sent to recycling. (10)



LEAKAGE IN THE SYSTEM

Today **over 60%** of the household waste is not sent to recycle after the waste management facility. A large part of this is due to errors made by the first line of recycling - the people at home - who makes sorting decisions and handles waste first-hand. When sorting and handling errors are made at this level, it creates major challenges for the recycling process. (11)



Norway's waste
14% Industries
24% Construction
21% Service Industry
21% Private Households
20% Unspecified

21% of all waste in Norway comes from private households. (12)

This waste actually weights the same as:

373 elephants

For those that don't speak "jungle", that means **2 419 000 tonnes** of private household waste every year.

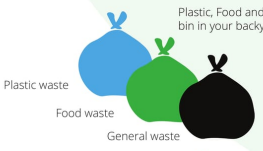
Oslo

THE CAPITOL: **OSLO**

Oslo, the major population center of Norway, uses the Optibag waste management system.

Optibag system
Blue for plastic, green for food and general waste in general plastic bags. You also sort paper and glass & metal.

Plastic, Food and General waste goes in the same bin in your backyard and Paper in its own bin. (14)



WASTE **MANAGEMENT** SYSTEMS

When the municipalities own the responsibility of recycling, many different waste management systems appear. The level of technology is different and the level of complexity for the user follows suit.

The following systems are used in Norway and this is the effective percentage on sorting plastic waste based on average plastic waste per person of 24kg a year:

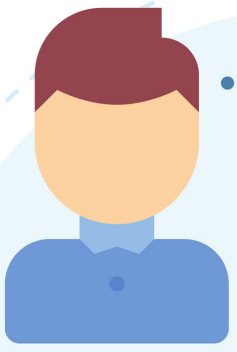
- Centralized sorting system 86 %
- Deliver sorting system 31 %
- Collecting sorting system 34 %
- **Optibag sorting 19 %**

(17)

Optibag sorting technology it is based on the color of the trash bag. This puts pressure on correct sorting and proper knowledge on the user. Example: if the blue plastic bag weights more than 600 grams it gets sorted to general waste no matter what. (15)

Centralized sorting is based on modern technology which scans the waste and sort out the plastic. Therefore it is less room for user error, which again results in higher percentage sorted out of the waste. (16)

FIRST LINE OF RECYCLING — PEOPLE



HOW DO I RECYCLE?

There are many factors that has to be taken under consideration when you are starting to recycle. It is not intuitively for people how this works, and it works in many different ways.

- First they need to know that you actually should recycle by law! There is no way around.
- You need to know what you should recycle based on where you live.
- You need to have a solution for recycling.
- You need to actually sort it when you get it in your home. If it's too hard, you won't do it.
- Therefore it is essential you get a good system to begin with.

37% of the people asked said they google when they are uncertain about how to recycle and only 18% say they use the information on the package. (23)

Kantar media did a big report on people barriers to recycling plastic. They found that people believe that the work associated with sorting, uncertainty about sorting and lack of good sorting solutions is the main reason for "people" not sorting plastic. (23)

People who don't sort plastic feels that there is no good solution in their municipality.

And they believe there is no use for sorting plastic since "they often believe it doesn't help" (23)

NEW RECYCLE **ICONS!**

To help making the recycling more understandable and easier for everyone to do, LOOP has created new symbols that are going to be implemented during 2020. They will start to be on the packaging as well as on the waste bins.



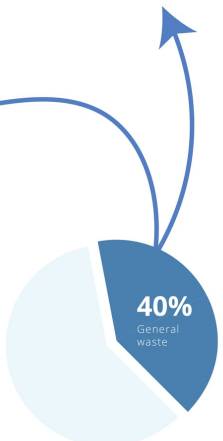
WHAT'S **IN THE BAG?**

A lot of things gets sorted wrong in general waste, 65,7% of the general waste is sorted incorrectly. This equals **54 857 tonnes** (2019) of waste that never gets recycled correctly.

The list on the left shows what the **general waste** consists of in Oslo municipality. Here food waste is the biggest part that is sorted incorrectly in general waste (36,3%), This is also the trend in the rest of Norway. (18)

FOOD	**36,3%**
GENERAL WASTE	26,6%
PAPER	9,0%
PLASTIC	8,6%
PET	6,1% (19)





WHY IS THIS **SO BAD?**

In almost all the municipalities in Norway general waste gets burned for electricity and remote heating.

Therefore, if you do not sort your waste, the materials might be lost forever.

By improving this we can recycle more materials and decrease the Co2 impact in the world. (21)

54,5%
Percentage of households waste is incinerated in Norway

WHAT **TYPE OF WASTE?**

When looking at the registered type of waste sorted in Norway, we see that 40% of the waste is from General waste. Thats a big concern when looking at what that waste group contains. And all general waste gets burned. (20)

